Lab 2 – Light or Paint? Intrinsic Images

CV8508 – Perceptual Cognition for Computer Vision

An image is the product of two things: the *reflectance* R of the surfaces (paint) and the *shading* S of the light, $I = R \cdot S$. Lecture 4 showed that people recover R from I using cues such as penumbra, ratios across edges and co-planarity. You will build two decompositions, one from a physical prior and one learned from human judgments in Intrinsic Images in the Wild [1], test both on Adelson's checker-shadow image and on test photos, and then ask a recent multimodal model to do the same. A coding agent is allowed, but check what it does.

Setup

```
mkdir lab2 && cd lab2
wget https://utkarshmall13.github.io/cv8508-labs/lab2/setup.sh
sh setup.sh                        # downloads data and lab2_tools.py, builds .venv
source .venv/bin/activate
```

Read `lab2_tools.py` first (loaders, WHDR, probe measurements; no decomposition code). `data/` then contains (each folder has a `README.txt`):

`iiw` a downsized IIW subset, `train/` and `test/`. Humans compared pairs of points in each photo: "same reflectance" or "one is darker", with a confidence weight. `load_iiw()` returns images and judgments. *Never train on `test`.*

`adelson` Adelson's checker-shadow image [2], with *probe* boxes on the squares A and B and the true albedo ratio of the two (`adelson.json`). The pixel values of A and B are almost the same (111 and 122 out of 255).

Tasks

Save every step as a script. Work in log space and in linear light.

1. **Look at the data.** Using a coding agent and `lab2_tools.py`, visualize the IIW images. Show a few training photos with annotated pairs drawn on them, coloured by label ("same" or "one point darker"), and note how many labels say "same".
2. **Method A: Retinex, no training data.** Implement Retinex [3] using the Retinex slides of Lecture 4 and the paper, and test it on the `test` images. The idea: a large step in log brightness between neighbouring pixels is paint, a small step is light. A recipe that works: take the log luminance $l = \log I$; look for a log shading s with $\log R = l - s$ that minimises $\sum |\nabla(l - s)| + \lambda \sum |\nabla s|^2$ (paint changes are few and sharp, shading is smooth), by gradient descent on s, solving a downsampled image first and using it as the start for the full size. Tune λ on `train` only and check convergence. (Keeping only log-gradients above a threshold and integrating is simpler but weak on photos.) Report the mean WHDR on `test`.
3. **Method B: a CNN trained on human judgments.** The network outputs a full shading map s for the whole image, with $\log R = \log I - s$ (start from $s = 0$), but IIW has labels only at about 170 point pairs per photo. So the loss uses only those points: for a pair (i, j) with confidence weight w, read $\log R$ at the two points; if humans said "same", the loss is $w\,|\log R_i - \log R_j|$; if they said "i is darker", it is $w \max(0, \log R_i - \log R_j + m)$ with a margin $m \approx 0.1$; and symmetrically for "j is darker". Add a small smoothness penalty on s. Only

the labelled pixels enter the loss, but the gradient reaches every weight through them, and a convolutional network shares its weights across the image, so it outputs a map everywhere. Keep it small (under about 500 000 parameters, at most 6 epochs). Report WHDR on `test`, next to Method A and next to the trivial answer "everything has the same reflectance". *Hint:* about two thirds of the judgments say "same", so a network that outputs log reflectance directly collapses to a constant; predicting the shading avoids this.

4. **The Adelson image and the test photos.** Run both methods on the checker-shadow image. Make one figure (image, A's, B's reflectance) and report with `probe_ratio` the three ratios listed in `adelson.json` (B/A, and the two pairs that have the same paint): in the pixels, in the truth, from Method A and from Method B. Compute B/A once in sRGB values and once in linear light. Then, on the `test` split, pick a few photos where A has the lower WHDR and a few where B does, and show each with both reflectances.
5. **A recent multimodal model.** Give the Adelson image and two of your test photos to a recent multimodal model that can return images (for example the Gemini app). Ask for an albedo image and a shading image, and ask in words whether squares A and B are the same shade of gray. Save what it returns, convert it to linear light, and measure the same probe ratios on the Adelson result. Record your prompts.

Analysis

Answer in a few sentences each, citing your own numbers, and mark each claim as evidence, hypothesis or speculation.

1. Using the presentation and your code, explain how WHDR works.
2. How do A and B compare on WHDR, and how do both compare with the trivial answer? With 300 test images, how large a gap would be meaningful?
3. On the Adelson image, what B/A ratio did each method recover (truth and pixels are in `adelson.json`)? How close is each to the truth, and why is the ratio in sRGB values different from the one in linear light?
4. What did each method report for the two pairs that have the same paint, and what does that say about how each treats shadows?
5. Which cue from the lecture does each method use, and which cue is missing?
6. Look at a few test photos and the outputs of A and B. Find cases where A works better than B and explain why; do the same for cases where B works better than A.
7. Did the multimodal model's words and its picture agree? What B/A ratio did its picture give, and how did it treat your photos? Could you tell it was wrong without the truth?
8. Propose one experiment on the Adelson image (for example changing the shadow edge) that would show whether a model uses shadow cues or just brightness context.

References

[1] Sean Bell, Kavita Bala, and Noah Snavely. Intrinsic images in the wild. *ACM Transactions on Graphics (SIGGRAPH)*, 33(4), 2014.

[2] Edward H. Adelson. Checker shadow illusion. https://persci.mit.edu/gallery/checkershadow.

[3] Edwin H. Land and John J. McCann. Lightness and retinex theory. *Journal of the Optical Society of America*, 61(1):1–11, 1971.